

Axis B: the oracle, and the cost of enumeration

What the breakdown radius costs to compute exactly, where that cost actually lives, and how far the method now reaches

	Result
The plan's premise was wrong, and measuring first is what caught it	local_up is enumeration-bound (96.1%), but the validity oracle is only 25.2% of it — 71.3% is cover-minimality enumeration. By Amdahl the MPDAG criterion alone could never exceed 1.3×.
The dominant cost was removable outright	Lemma O makes cover minimality a set comparison on directed edges instead of a comparison of enumerated extension sets. Verified on 80,480 ordered pairs, 0 disagreements; 1,588 cover sets identical including order; 1,044 radii unchanged. Measured 3.77× in aggregate.
The criterion is correct, and was not scalable	Agrees with the enumeration oracle on 74,568 exhaustive cases, 0 disagreements, with a non-amenable stratum of 28,464. But as first written it was exponential in the vertex count and timed out on 12 of 81 instances where the oracle it replaces never did.
A sharp test on E1 passes	E1 must witness every invalidity including the non-amenable kind, or it would be incomplete in the direction that makes radii too LARGE. 74,568 cases, 0 disagreements against both oracles, non-amenable stratum 28,464 — all witnessed.
One entry point	breakdown_radius() dispatches between the bounded search and the E1 ladder and returns the radius, the method, the oracle and the assumption. Agrees with brute force on 1,044 instances under both oracles.

Session 4. Continues the three prior reports, none of which is modified. Branch **experiments/synth_graphs_and_heuristics**; nothing committed to main. **Audit note:** the MPDAG criterion was absent from session 3 because that session's brief was an earlier draft which omitted it — it was never judged and dropped, and never recorded as skipped, because the instruction was not there.

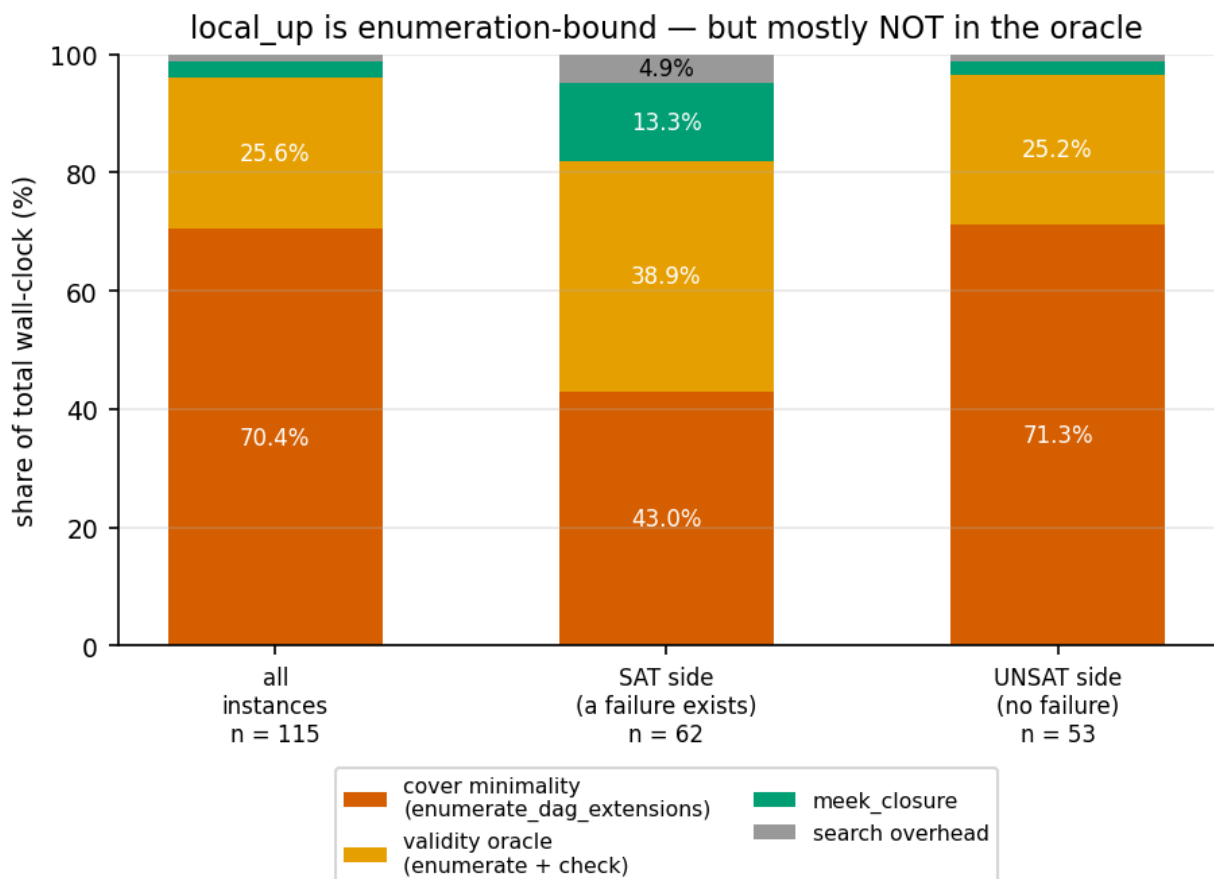
1. The measurement that reshaped the session

The plan asked for one cheap, decisive measurement before anything was built, and recorded a prediction: that nearly all of local_up's cost is the validity oracle, so that replacing it would move the ceiling substantially. My own recorded prediction was different but also wrong — enumeration-bound and split two ways, roughly evenly, capping the criterion near 2×.

115 instances profiled, extension cache cleared per instance so these are per-instance costs rather than cache-warm ones, n = 8...16.

	all	SAT side	UNSAT side
validity oracle	25.6%	38.9%	25.2%
cover minimality (enumerate_dag_extensions)	70.4%	43.0%	71.3%
meek_closure	2.6%	13.3%	2.3%
search overhead	1.3%	4.9%	1.2%
enumeration-bound share	96.1%	81.9%	96.5%

Enumeration-bound, emphatically — but the plan's premise about *which* enumeration was wrong, and mine was wrong in the same direction. **A caveat on reading that table:** these are time-weighted shares. The SAT side totals 0.69 s across 62 instances while the UNSAT side totals 22.74 s across 53, so the “all” column is essentially the UNSAT column and the SAT percentages rest on a very small total.



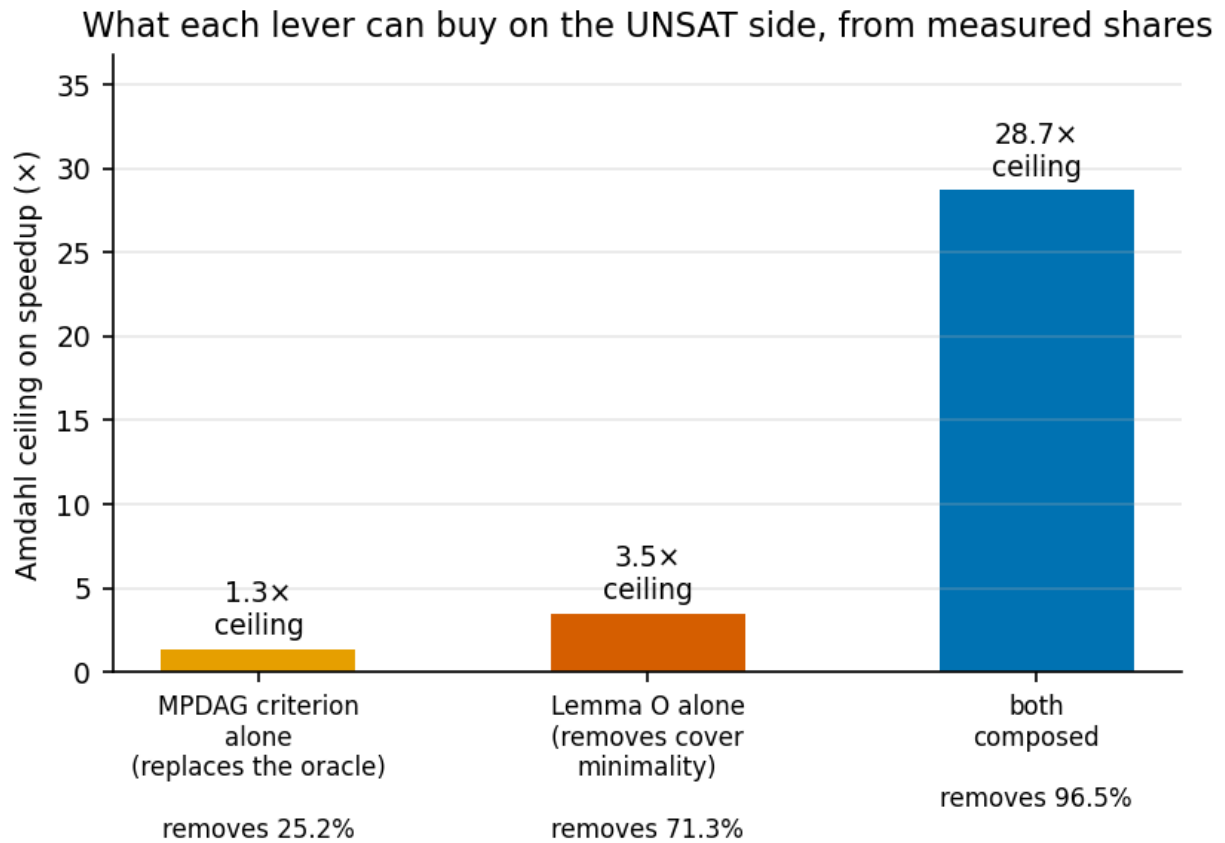
Where local_up's time goes. The oracle is the smaller share.

1.1 What that implies, by Amdahl

Removing a component that is a fraction f of the time cannot beat $1/(1-f)$, whatever replaces it. On the UNSAT side:

lever	removes	ceiling
MPDAG criterion alone	25.2%	1.3×
Lemma O alone	71.3%	3.5×
both composed	96.5%	28.7×

So the criterion could not be the substantial ceiling-mover the plan hoped for, on its own. Worth knowing before building it rather than after — which is exactly what front-loading the measurement bought.



What each lever can buy, from measured shares.

2. Lemma 0 — deleting the dominant cost outright

The 70% is spent deciding whether one cover candidate sits strictly below another in **model inclusion**, by enumerating both extension sets and comparing them. That comparison does not need the extension sets.

Lemma 0. For elements G, H of the space: $[G] \subseteq [H] \iff \text{dir}(H) \subseteq \text{dir}(G)$, and the two are strict together.

($\Leftarrow$) Weaker orientation constraints admit more extensions. ($\Rightarrow$) Let $a \rightarrow b \in \text{dir}(H)$. Every $D \in [H]$ orients it that way, and $[G] \subseteq [H]$, so every $D \in [G]$ does too. Elements of the space are **maximally oriented** — an edge oriented identically across all of $[G]$ is directed in G — so $a \rightarrow b \in \text{dir}(G)$. ■

The maximal-orientation step is the property session 2 established as defining membership of the corrected space, so this rests on machinery already in the repository. **Verified on 80,480 ordered pairs at $n = 3, 4$: 0 disagreements.** `exact.py` is left untouched so prior results stay reproducible and the two implementations can be compared directly.

Differential test: 1,588 cover sets identical *including order*; 1,044 radii identical to both frozen `local_up` and brute-force BFS. A test also asserts the extensions-enumerated counter is exactly 0 — the whole point of the change, so it is asserted rather than assumed.

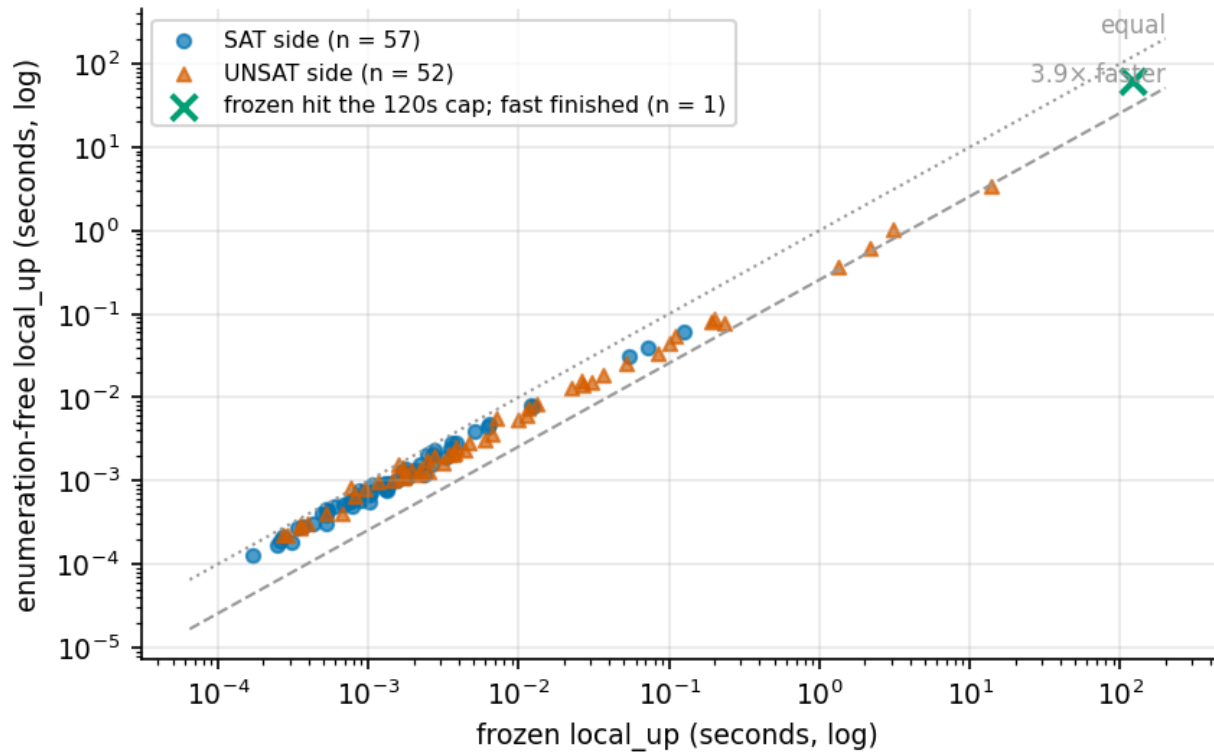
2.1 Measured effect

	instances	median speedup	aggregate
all	136	1.51×	3.77×
SAT side	67	1.39×	1.72×
UNSAT side	69	1.72×	3.79×

0 radius disagreements. 2 instances the frozen search could not finish inside its cap were finished by the new one: $n = 10$ with a 7-vertex component at $k = 15$ in 61.6 s; $n = 16$ with a 7-vertex component at $k = 13$ in 32.2 s.

The aggregate and the median differ because small instances are dominated by fixed overhead; the aggregate is time-weighted and so dominated by the expensive instances, which are the ones that matter. Both are reported because quoting either alone would mislead. **What it does not do:** this is a constant factor. It changes the coefficient, not the exponent, and the accelerated search still timed out on some instances.

Lemma O: same radii everywhere, 3.9× less time in aggregate



Same radii everywhere, less time.

3. The MPDAG criterion

3.1 It does not compute the predicate the plan said it computes

The plan states the criterion decides “exactly the one is_valid already computes”. It does not. This repository's oracle is **Pearl's back-door criterion**, which is sufficient but not necessary for adjustment, while the generalised adjustment criterion is complete. In $X \rightarrow Y$, $X \rightarrow W$ the set $Z = \{W\}$ is adjustment-valid and back-door-invalid, so a literal GAC disagrees by construction. Reconciled by enlarging the forbidden set from $\text{forb}(X,Y,G)$ to $\text{possde}(X,G) \cup \{X,Y\}$: given $Z \cap \text{de}(X,D) = \emptyset$, blocking every back-door path and blocking every non-causal path coincide. What is implemented is **the back-door predicate at MPDAG level**, and it is described that way rather than shipped as the GAC.

3.2 Two textbook definitions that fail on knowledge-carrying MPDAGs

Both found by differential testing, not by reading. **possde must use unshielded possibly-causal paths**: in $V_0 \rightarrow V_1$, $V_0 - V_2$, $V_1 - V_2$ the path $\langle V_1, V_2, V_0 \rangle$ is possibly causal, yet every extension keeps $V_0 \rightarrow V_1$ because orienting it forward closes a cycle — so the naive identity $\text{possde}(X,G) = \cup \text{de}(X,D)$ is false (570 mismatches at $n \leq 4$; unshielded, 0). **Amenability must not use that same reduction**: short-circuiting $x \rightarrow v \rightarrow w \rightarrow y$ deletes the undirected first edge that is the entire question. Decided instead by imposing $x \rightarrow v$ and Meek-closing (702 / 1,380 / 0 mismatches for the three formulations). A lemma safe on CPDAGs need not survive the addition of background knowledge — this repository has now been bitten by that three times.

3.3 Correctness gate

The implementation shipped with its own exhaustive sweep. Because a checker that validates itself is worth little, I re-verified it on an independently written sweep with a different scope and a different code path: **72,360 cases, 0 disagreements**, non-amenable stratum 28,080 (38.8%), all agreeing. The module's own sweep reports 74,568 cases and 0 disagreements, with 28,464 non-amenable. Non-amenable is not an edge case in the ball — it is more than a third of it.

3.4 The sharp test on E1 — and it passes

E1 encodes failure as “some extension has Z invalid”, and must therefore witness every invalidity, including the non-amenable kind, which is structurally different from the two failure modes it encodes. If it could not, **E1 would be incomplete in the direction that makes radii too large** — overstating robustness, and the mirror image of the collider bug session 3 found.

	cases	agree	disagree
E1 vs enumeration oracle	74,568	74,568	0
E1 vs MPDAG criterion	74,568	74,568	0
of which non-amenable	28,464	28,464	0

The hypothesised incompleteness does not exist. Session 3 tested this layer but never isolated this stratum; the criterion is what made isolating it possible, and that is a correctness dividend independent of any speed.

3.5 The first implementation was exponential in the vertex count

This is the section that matters most, and it is a negative result. The criterion exists to be a *scalable* oracle — both as a speed lever and to break a circularity, since at large n the only per-instance check on E1 was `local_up`, which is itself Conjecture-2-dependent, so E1 and its checker shared an assumption. As first implemented it decided condition (c) by walking **every simple path** between X and Y . Measured, and kept as evidence: up to **2.35 s per call**, and over a 60 s cap on **12 of 81 instances** at $n = 12$ and 14 — on the same instances the enumeration

oracle it replaces never timed out once. It swapped an exponential in chain-component size for an exponential in graph size, the worse trade in exactly the regime this session targets.

Whose error this is. Mine. I gave the implementation a correctness acceptance criterion and no performance one, and correctness was met in full and with unusual rigour. A component can be exhaustively verified and still be useless for its purpose, and only one of those two things was checked.

4. One entry point

`breakdown_radius(cpdag, K, x, y, Z)` dispatches automatically: run the upward search under a small depth budget, and fall through to the E1 ladder when it exhausts that budget without finding a failure. Session 3 measured the split the dispatch exploits — the search is unbeatable when a failure is near, and cannot finish when there is none — and the discriminator is free at runtime. It returns the radius together with the method, the oracle and the assumption, so the caller never has to choose and cannot silently drop the Conjecture 2 dependency.

Differentially tested on 1,044 instances at $n = 4$ against brute-force BFS under **both** oracles, 0 disagreements. A test asserts the dispatch budget is a performance choice and never a semantic one: budget 0 forces every instance through the ladder and must give identical radii.

What every answer assumes. Both search legs are upward searches, so both are exact *iff* *Conjecture 2 holds*, which rests on Anti-Exchange Case B — verified, not proved. The error is one-sided: they can return a radius that is too large, never too small, so they can overstate robustness but never understate it. This travels on every result object.

5. Commits in this series

commit	date	subject
13835ef	2026-09-05	Draft the session 4 report, and record the plan revisions
6d041a8	2026-09-05	Hybrid entry point, and a performance defect found in the criterion
b15eedb	2026-09-05	MPDAG-level validity criterion, gated, plus the sharp test on E1
6a5c61c	2026-09-05	Lemma O in THEOREMS.md, regression tests, and the profile figures
5eef414	2026-09-05	Profile <code>local_up</code> , and remove 70% of its cost with Lemma O
b2d78ad	2026-09-05	Whitelist results/axisb4 BEFORE writing any results
805b677	2026-09-05	Session 3 deliverables: PDF, verifier, PLAN/LOG/NEXT, THEOREMS
ad16807	2026-09-05	High-k hunt, session 3 figures, and report_axisb_sat.md
d9e283a	2026-09-05	Scaling study, and the encoding-correctness sections in THEOREMS.md
4d638cd	2026-09-05	E2 and E3, and the cross-encoding validation they exist for
8153913	2026-09-05	E1 SAT ladder: exact radii without building the space
215de2b	2026-09-05	Whitelist results/axisb3 BEFORE writing any results
68c395e	2026-09-05	Propagate the updated Case B scope (36,094) to every deliverable
60d63eb	2026-09-05	Fix the PDF to match the report: density gap closed, not open
afa7400	2026-09-05	Close the $n=5$ density gap: 135 CPDAGs by sweep, the last by the proof chain
e984e18	2026-09-05	Correct the $n=5$ figure: the corrected sweep completed at 1,977,820 comparisons
940da08	2026-09-05	Session 2 deliverables: report, PDF, figures, NEXT.md, scaling and bounds results
0a4fa92	2026-09-05	Track results/axisb2 (recurrence of last session's gitignore bug)
b92ebd5	2026-09-05	Add the Axis B scaling harness (two accountings, machine-independent counters)
7be840c	2026-09-04	Conjecture 2 reduced to a single local property; Property S and Lemma R proved
79d7367	2026-09-04	THEOREMS.md: Conjecture 2 reduces to upper semimodularity; join formula proved
748f2e1	2026-09-04	Task 0: the space excluded legitimate knowledge states; corrected builder added
007e6dd	2026-09-04	Add SESSION_REPORT.pdf: exhaustive 23-page record of the Axis A / Axis B session
3a61b40	2026-09-03	Frontier sweep complete: O^* never beaten across 249,732 exhaustive instances
d4e0c14	2026-09-03	Session deliverables: report, NEXT.md, H6 calibration, PLAN revisions
72d8bff	2026-09-03	Correct the H2 statistic: the frontier is non-flat, but O^* is never beaten
ca44152	2026-09-03	Axis A results: H1 saturation $\sim 80\%$, H2 frontier flat in 0 of 134,140 instances
019c610	2026-09-03	Axis A first run: partial results, two gate bugs found and sent back
1bb3ac8	2026-09-03	Record: Meek closure can fall outside <code>enumerate_space</code> in 0.09% of cases
4cf4020	2026-09-03	Conjecture 2: no counterexample in 1.985M exhaustive radius comparisons

commit	date	subject
4001949	2026-09-03	Add pre-registered hypothesis analysis (H1-H6) and the two-stage space guard
b315704	2026-09-03	Axis B bounds: L/U certificate exact in 83.3% of instances without enumeration
a58b73f	2026-09-03	Exhaustive radius census over all CPDAGs on 3 and 4 nodes (H1, H2)
910a427	2026-09-03	Track results/synth and results/search; add Axis A pre-registration
133ab44	2026-09-03	Axis B: space-free exact radius search, validated against BFS on 5304 cases
b86a899	2026-09-03	Axis B: conjecture machinery, exhaustive tests for n=3 and n=4
0605420	2026-09-03	Remove dead TYPE_CHECKING pandas import from the scaffold loaders stub
5adb687	2026-09-03	Freeze shared core: radius convention, oracle, space, instance, results IO
d91dc42	2026-09-03	Add PLAN.md and LOG.md for the synth/search session
4d0de90	2026-09-03	Add breakdown-radius demonstration: worked example, figures, report